

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Computer Vision Research Engineer

Computer Vision Research Engineer with 3+ years researching, tuning and deploying vision systems across safety-critical industries for clients including BP and Austin Industries. Experienced researching and benchmarking current-generation vision models (YOLO, RT-DETR, MobileSAM, Depth Anything v2) to guide model selection and deployment trade-offs, alongside adversarial training to improve accuracy and robustness under occlusion, lighting change, and motion artefacts. Skilled in Python, OpenCV, PyTorch, TensorFlow, Docker and Jira, and shipping production code alongside platform engineers.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Built benchmarking and profiling harnesses to evaluate models (YOLOv5, EfficientDet) across accuracy and latency post-quantisation alongside model-promotion gates to prevent quality regression.
- Designed adversarial-condition training and validation experiments (occlusion, lighting variation, motion artefacts) to benchmark robustness via F1 and failure-rate metrics, improving F1 by 15% while maintaining deterministic 24/7 operation on BP rigs.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts reducing live video pipeline crashes by 67% and improved root-cause analysis.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.
- Built synthetic image generation and augmentation (OpenCV, scikit-image, albumentations) to improve model robustness reducing false positives and supporting long-term reliability.
- Managed sprints via Jira coordinating deliverables across multidisciplinary teams and presenting model performance and trade-offs to engineering, client, and field stakeholders.

livSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy vision pipelines for real-time collision avoidance on live construction sites in Texas for Austin Industries.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing robustness.
- Applied Grad-CAM and D-RISE for model interpretability and TIDE-based error taxonomy for failure analysis to diagnose recurring detection failures, prioritise dataset fixes, and reduce false positives by 33% across camera feeds.
- Mentored junior engineers and collaborated with hardware, software, QA, and client teams to integrate perception modules and translate performance, latency, and failure analysis into rollout decisions for deployed safety systems.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed a real-time SLAM system on a Clearpath Husky using RTAB-Map, ROS2, and C++17, integrating LiDAR and radiological data for inspection-scenario localisation and mapping.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

PROJECTS

DeepStream RTSP Pipeline

Jun 2026

- Built an end-to-end NVIDIA DeepStream 9.0/GStreamer RTSP video analytics pipeline for multi-stream computer vision, using rtspsrc, nvstreammux, nvinfer, nvtracker, and nvdssd to run YOLO+NvDCF tracking, metadata extraction, anonymisation, CSV logging, and re-streamed output across 3 concurrent camera feeds.
- Deployed and optimised a YOLO26n TensorRT FP16 detector on 6GB VRAM Turing hardware, combining dynamic-batch engine conversion, nvinfer integration, p99 latency profiling, and FP16-vs-FP32 accuracy validation to sustain 29.7 FPS per stream live and 131.3 FPS per stream unthrottled.
- Implemented a C++/CUDA TensorRT decode plugin using IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Added production hardening through GPU smoke tests, model-promotion gates, structured JSON logging, and per-camera health metrics for reactive debugging across liveness, dropped frames, reconnect events, and RSS.

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, YOLO26, RT-DETR, EfficientViT DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.
- Exported models to ONNX and evaluated OpenVINO deployment paths, strengthening practical understanding of model optimisation, inference acceleration, and edge deployment constraints.
- Built benchmark comparisons using mAP, inference time, and model-size measurements to support model selection for low-latency visual AI systems.

SKILLS

- **Computer Vision & Machine Learning:** Convolutional Neural Networks, GRAD-CAM, NumPy, OpenCV, PyTorch, Scikit-Image, Scikit-Learn (sklearn), TensorFlow, TIDE (Toolbox for Identifying Detection Errors)
- **Edge Inference and Optimisation:** custom TensorRT plugins, FP16/INT8 deployment, latency/throughput profiling, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, Quantisation-Aware Training, Structured pruning
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, Docker, DVC (Data Version Control), FastAPI, Git, Health checks, Intel VPUs, Linux, Microsoft Azure, MLflow, model-promotion gates
- **Programming & Tooling:** Bash, C++17, CI/CD, Pandas, PostgreSQL, Python
- **Edge AI & Video Pipelines:** camera behaviour debugging, CUDA, FFmpeg, GStreamer, multi-stream video analytics, NVIDIA DeepStream 6.x/9.0, TensorRT